Simulation and characterisation of a chaotic oscillator using LTspice

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Abstract – This report investigates the behaviour of Chua's oscillator circuit using LTspice simulations, focusing on visualising how changes in circuit parameters influence the transition from weak oscillations to fully developed chaos. Figures illustrate the effects of component values on the chaos. Through iterative tuning of RLC values and circuit configuration, the circuit progresses from stable oscillations to the distinctive chaotic double-scroll attractors. The results emphasise the circuit's high sensitivity to small parameter variations, demonstrating the fundamental principles of non-linear and chaotic systems.

I. Introduction

A chaotic oscillator is a type of electronic circuit that produces a complex, non-repeating signal that may appear random, however, in theory can be entirely determined by equations and component properties. An example of this is Chua's circuit shown in Fig. 1. It contains two capacitors and an inductor that store and exchange energy, which is coupled to the non-linear Chua's diode. Chua's circuit exhibits chaotic dynamics due to its sensitive behaviour on initial conditions.

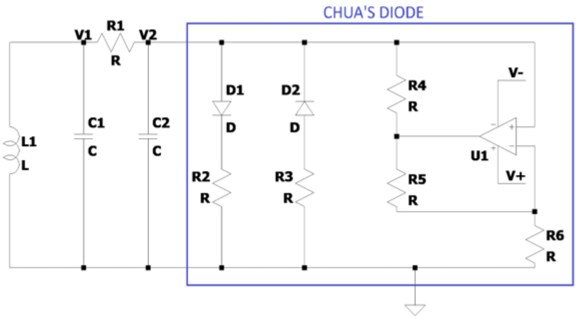

Fig. 1: Schematic of the basic Chua oscillator circuit. Chua's diode highlighted in blue is constructed using the following: 2 diodes (D1, D2), 5 resistors (R2-6) and an op-amp (U1) powered by voltages V^+ and V^-, which has a piecewise linear characteristic. Chua's diode creates a non-linear effect that causes the system to behave in a highly sensitive and unpredictable manner. Chaos in this circuit is displayed by plotting the potential difference V1 and V2 across the two capacitors C1 and C2. When V1 is plotted against V2, in theory, the oscillations should turn into a double scroll pattern [1].

II. Methodology

LTspice was used to simulate Chua's circuit and explore how different component values influence chaotic behaviour. Initial component values were selected from realistic ranges for resistors, capacitors, and inductors, and standard 1N4148 diodes were used to form the nonlinear Chua diode. The OP07 op-amp was first tested and later replaced with the LT1113 to improve precision and noise performance. A transient, time-domain simulations were run with a 200ms stop time and a 0.1ms time-step to capture fast voltage oscillations. The RLC values were then systematically varied to observe how the circuit transitions from stable behaviour to chaos. Through iterative tuning, clear chaotic waveforms, including single and double scroll attractors were obtained and the successful component values were recorded for further refinement.

III. Results

For the initial circuit, the following configuration was selected: L1=18mH, C1=10nF, C2=10nF, R1=1.6kΩ, R2=22kΩ, R3=22kΩ, R4=3.3kΩ, R5=39kΩ and R6=3.9kΩ, Op-amp model: OP07, Power supply: 12V DC. V1 and V2 is probed across capacitors C1 and C2 against time, a chaotic waveform was created, as shown in Fig. 2. V1 and V2 is then plotted against each other to observe and visualise the state-space trajectory of the system, as shown in Fig. 3.

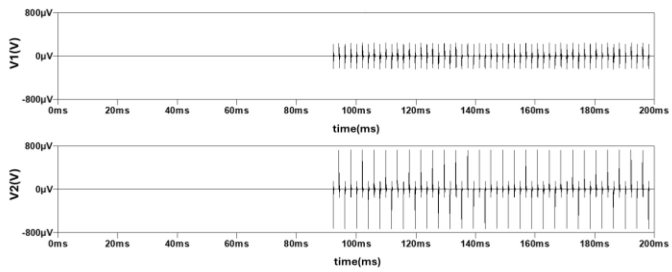

Fig. 2: Oscillation of V1 and V2 in the time domain of the initial circuit. The amplitude of the voltage oscillation of the Chua's circuit is very small in amplitude, seen as small spikes peaking at 200µV for V1 and around 800µV for V2. The system also takes around 90ms to develop, which is a significant time. This indicates that there is heavy damping trapping the system near a stable point. The arrangement of Chua's diode could be a potential cause as the diode does not provide enough non-linearity and negative resistivity for the system to oscillate.

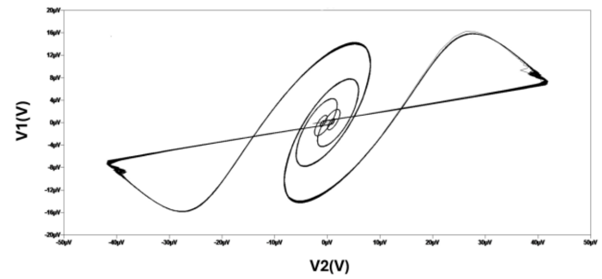

Fig. 3: Plot of V1 against V2 in initial circuit. It shows a single scroll attractor showing limited chaotic behaviour. The trajectory formed remains confined to one region and small in amplitude. This implies the system never switches between the positive and negative resistance regions in Chua's diode. The very small amplitude suggests that the circuit remains close to a stable operating point, meaning the non-linear element is not being driven strongly enough to produce a double-scroll attractor. This demonstrates the initial behaviour of the system and highlights the need for parameter adjustments to achieve full chaos, specifically the capacitor values.

To enhance the non-linearity of Chua's diode, some of the circuit components are modified by an iterative process. The modified circuit is set in the following configuration: L1=10mH, C1=50nF. C2=5nF, R1=1.7kΩ, R2=2kΩ, R3=2kΩ, R4=1kΩ, R5=1kΩ and R6=1kΩ, Op-amp model: LT1113, Power supply: Dual ±12V DC. A schematic of the modified circuit is shown in Fig. 4. Graphical results are shown in Fig. 5 and Fig. 6.

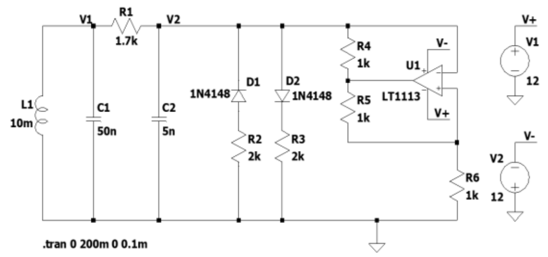

Fig. 4: Modified Chua's circuit. This figure displays an improved circuit schematic and fine-tuned parameter values for Chua's circuit. The op-amp model used is switched to LT1113, giving lower noise and higher precision properties. An additional power supply is added to give the op-amp true ±12DC rails, where each positive and the negative voltages are grounded. This reduces ambiguity and enhances voltage swings. It also ensures that the non-linear Chua's diode is correctly biased. The RLC values selected allow V1 and V2 traverse across the whole range of voltage in Chua's diode.

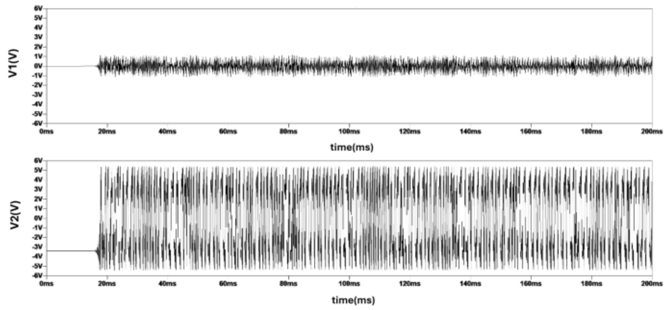

Fig. 5: Waveform of V1 and V2 in modified circuit in the time domain using previous simulation settings. Using the modified system, large irregular aperiodic voltage oscillations on a volt-scale across C1 and C2 is achieved. Amplitude of oscillations is around 1.2V in C1 and around 5.5V in C2, which is more significant than the initial circuit. Oscillations start as early as 20ms, whereas in the first simulation it took more than 80ms. This proves the degree of damping has been reduced, which prevents the system to be confined in steady state.

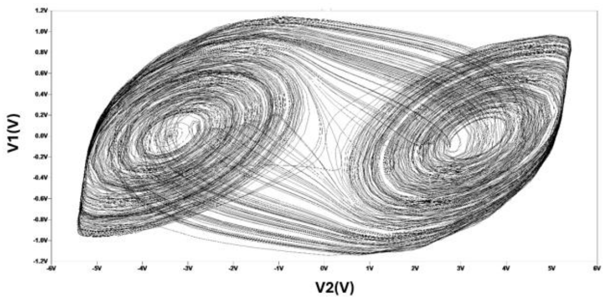

Fig. 6: Plot of V1 against V2 for the improved circuit. It shows V1 and V2 oscillating around two distinct equilibrium points, forming a double scroll attractor. The system does not follow a single closed path like a periodic oscillator, but instead moves unpredictably between two lobes, never repeating the same trajectory, demonstrating aperiodic motion. The range of voltage values shown are also now bigger in amplitude. By editing the configuration of Chua's diode, the system is made to be more non-linear and more chaotic.

IV. Discussion

I-V Characteristics: To investigate the I-V characteristic of Chua's diode, an I-V graph was plotted using the configuration of the diode shown in Fig. 4. A separate schematic was used to separate the diode with the main circuit elements. The diode was connected in series to voltage source V_{test}, where V_{test} is varied from -12V to 12V using the component value sweeping function in LTspice. The resultant diagram is shown in Fig. 7.

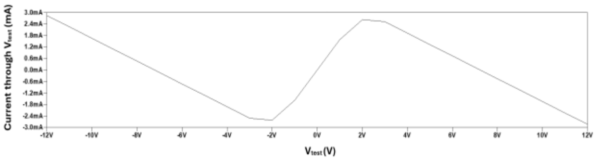

Fig. 7: I-V Characteristic of Chua's diode under the configuration in Fig. 4. The graph shows a 5-segment piecewise function of current against voltage. Instead of showing negative resistance throughout, the plot exhibits a shallow positive gradient near 0V and steep negative gradients in the outer regions, which matches with historical experimental results [2]. The circuit achieves chaotic behaviour because the effective feedback through the RLC network produces overall energy injection and dissipation required for chaos. This demonstrates that a piecewise-non-linear response and breakpoint voltage symmetry of Chua's diode are a sufficient requirement for Chua's circuit.

Capacitor Ratios: The capacitors C1 and C2 in a Chua oscillator circuit play a critical role in defining chaos. For the creation of the graph shown in Fig. 6, the ratio of C1 to C2 was 10. To see the effect of capacitance, the ratio was changed to 13.33, using C1=40nF and C2=3nF.

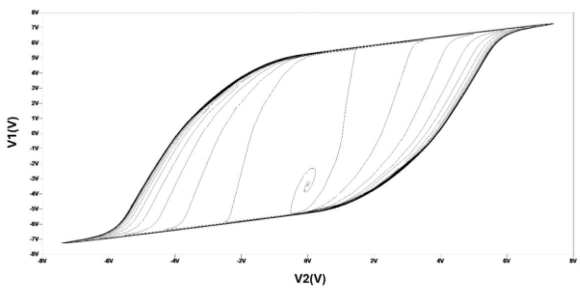

Fig. 8: V1 against V2 with altered capacitance values. The result was as expected; the graph was still chaotic but instead of a double-scroll there is one, as seen in Fig. 8. This means, the chaos has now decreased, and the graph is now more stable. Fig. 8 shows that even a slight derivation from a capacitance ratio of 10 reduces the chaotic feature of the oscillator significantly, as the capacitor differences alter the energy exchange rate between them, weakening the nonlinear interaction required for double-scroll. Chua's oscillator gives optimal chaos when a capacitance ratio of 10 is used.

Chua's Diode parameters: After multiple iterations, the final configuration of the circuit with R2 and R3 set to a one-to-one ratio. The process of reaching this point required tuning the resistors that define the diode's non-linear region. Fig. 9 shows a waveform where the resistor ratio is slightly adjusted.

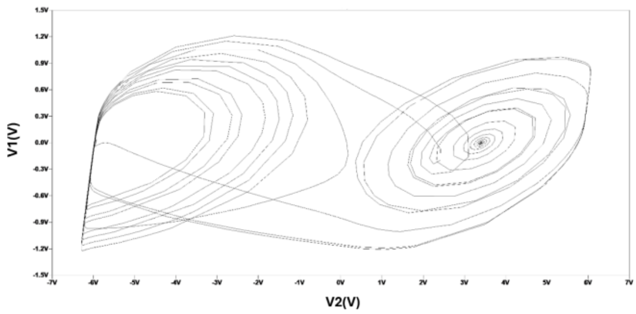

Fig. 9: Plot of V1 against V2 with R2 altered to 2.2kΩ. This shows a collapse of the double scroll image, demonstrating the unique sensitivity of Chua's diode. Even a small change in resistance produced a large alteration in the circuit behaviour and results. This demonstrates the extreme sensitivity of Chua's circuit to component values, a key feature of chaotic systems. The non-linear I–V characteristic of the diode means that tiny variations can push the circuit between stable, periodic, or chaotic states. This shows the final parameters achieved a balanced region, where energy exchanges between components ensures continuous chaotic oscillations.

A. References

[1] J. Eric, *"LTspice Simulation of Chua's Circuit,"* Blogspot.com, Dec. 2011. Accessed: Oct. 24, 2025. [Online]. Available: https://juho-eric.blogspot.com/2011/12/ltspice-simulation-of-chuas-circuit.html\

[2] *G.-Q. Zhong and F. Ayrom, "Periodicity and chaos in Chua's circuit," IEEE Transactions on Circuits and Systems, vol. 32, no. 5, pp. 501-515, May 1985.*